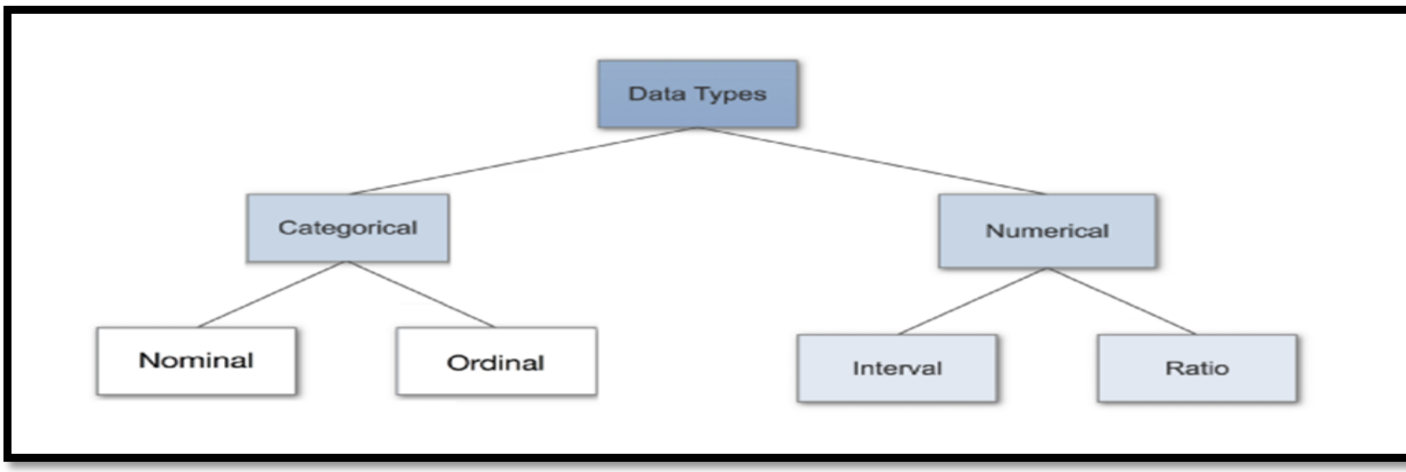
Internship Report

In these 30 days of internship, we understood the concepts of measurement of data, descriptive statistics/ inferential statistics with conceptual clarity and with the real time applications.

DATA:

We learnt about the measurement of scales such as Nominal, Ordinal, interval and Ratio scales with examples



Scale	Characteristics	Examples
Nominal	•Label and categorize •No quantitative distinctions	•Gender •Diagnosis •Experimental or Control
Ordinal	•Categorizes observations •Categories organized by size or magnitude	•Rank in class •Clothing sizes (S,M,L,XL) •Olympic medals
Interval	•Ordered categories •Interval between categories of equal size •Arbitrary or absent zero point	•Temperature •IQ •Golf scores (above/below par)
Ratio	•Ordered categories •Equal interval between categories •Absolute zero point	•Number of correct answers •Time to complete task •Gain in height since last year

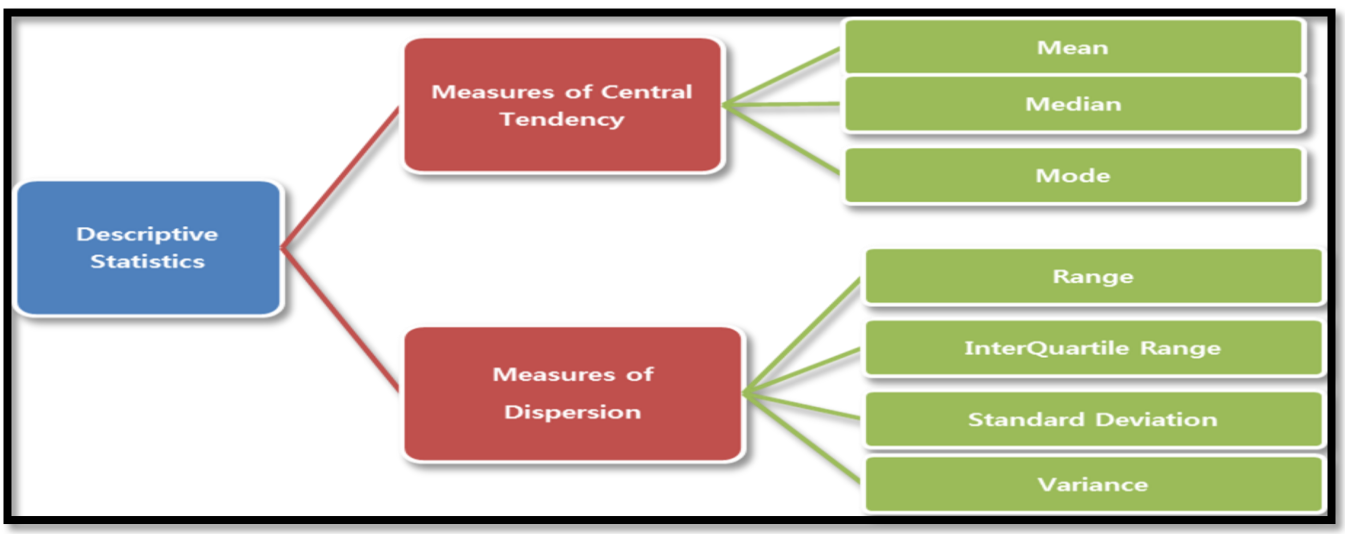
variable	Scoring	Nominal	Ordinal	Contin.
Quality of life	1 = Poor 2 = Fair 3 = Average 4 = Good 5 = Very Good		●	
Ethnicity	1 = Non-Hispanic 2 = Hispanic	●		
Race	1 = African American 2 = Caucasian 3 = Other	●		
Diabetes	1 = Absent 2 = Present	●		
Systolic BP	Ranges from 95 to 190 mmHg			●

DESCRIPTIVE STATISTICS:

We understood the concepts of measures of central tendency and measures of dispersions, Skewness and kurtosis along with their applications in real time.

MEASURES OF CENTRAL TENDENCY	*Location*
MEASURES of DISPERSION	*Spread*
SKEWNESS	*Shape*
KURTOSIS	*Height*

And also learnt about deciles, Percentiles, Quartiles etc..



- *MEAN is not applied when there are extreme values.*
- *MEDIAN is applied when there are extreme values.*
- *MODE is used for finding the most frequently occurring value.*

INFERENTIAL STATISTICS:

Inference is the process of drawing conclusions or making decisions about a population based on sample results.

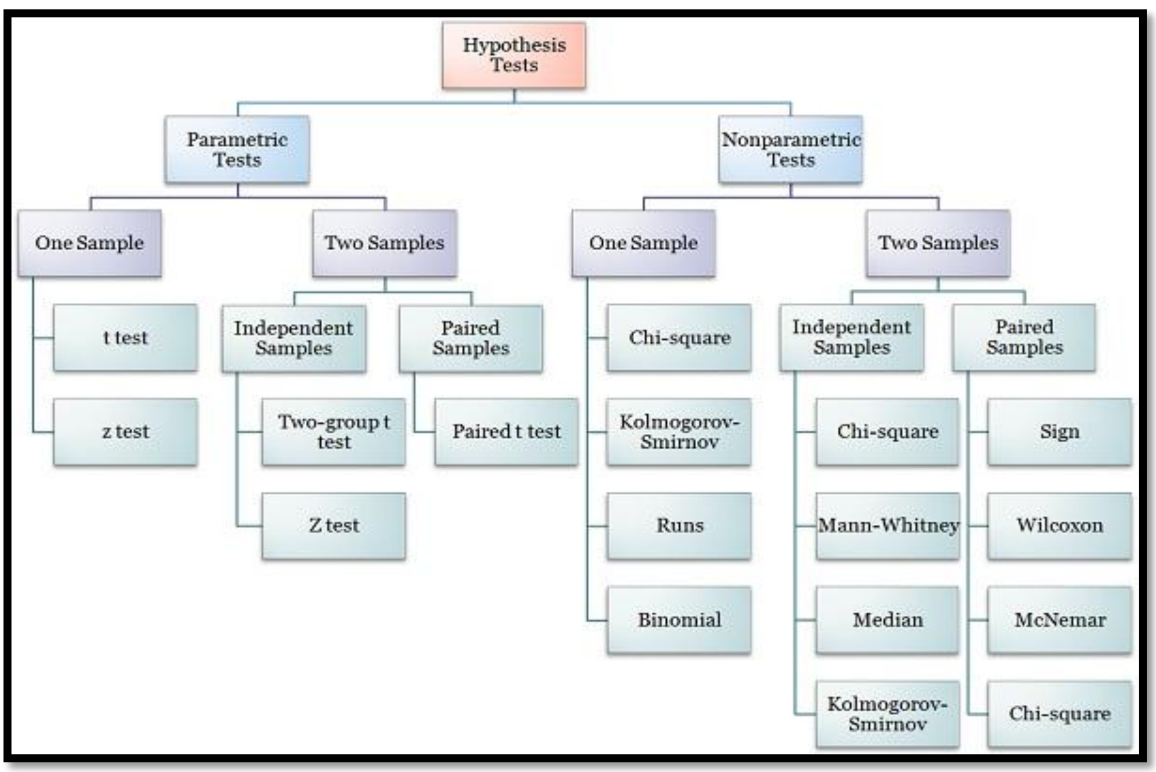
- **Estimation**
  - e.g., Estimate the population mean weight using the sample mean weight
- **Hypothesis testing**

Hypothesis is also called *significance testing.*

Tests a claim about a parameter using evidence (data in a sample)

e.g., Test the claim that the population mean weight is 70 kg.

- **Hypothesis Testing Steps**
  - A. Null and alternative hypotheses
  - B. Test statistic
  - C. P-value and interpretation
  - D. Significance level (optional)

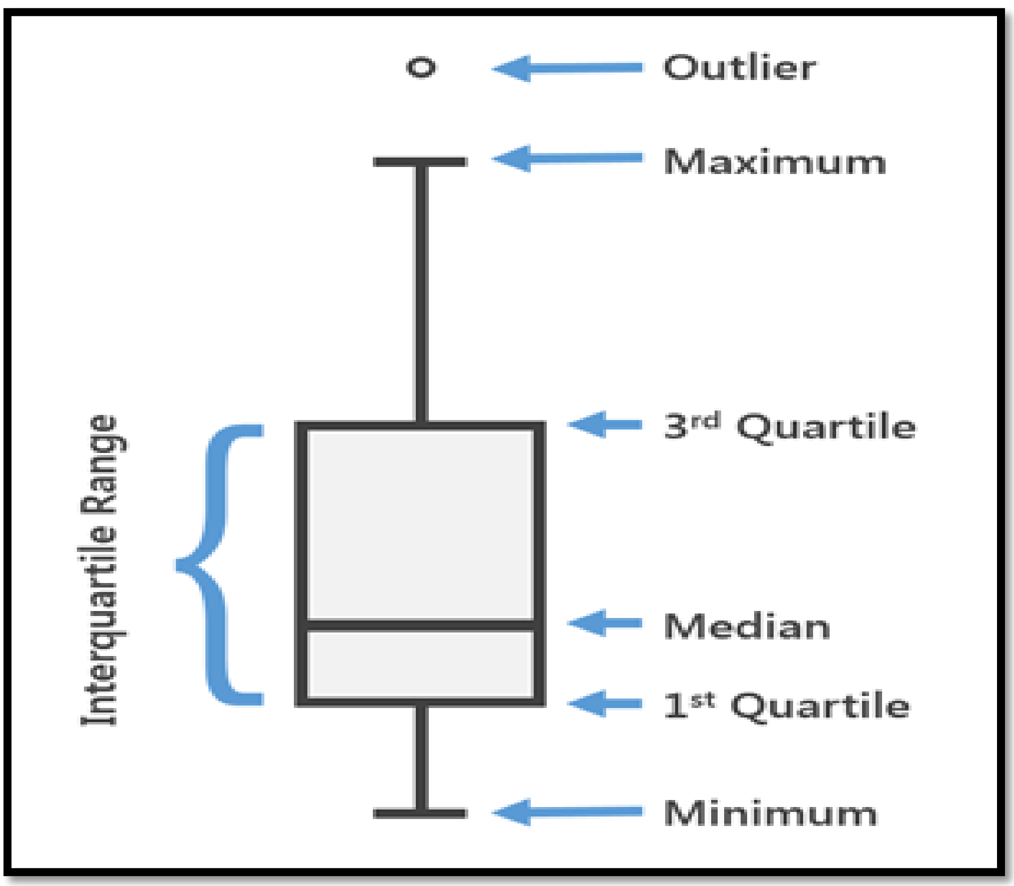


FEATURES/RESPONSE	CONTINUOUS	CATEGORICAL
CONTINUOUS	PEARSON'S CORRELATION	T-TEST/ANOVA
CATEGORICAL	ANOVA	CHI-SQUARE

BOXPLOT:

A box plot is a graphical rendition of **statistical** data based on the minimum, first **quartile**, median, third **quartile**, and maximum. The term "box plot" comes from the fact that the graph looks like a rectangle with lines extending from the top and bottom.

- Boxplot is used to find the outliers (extreme values).



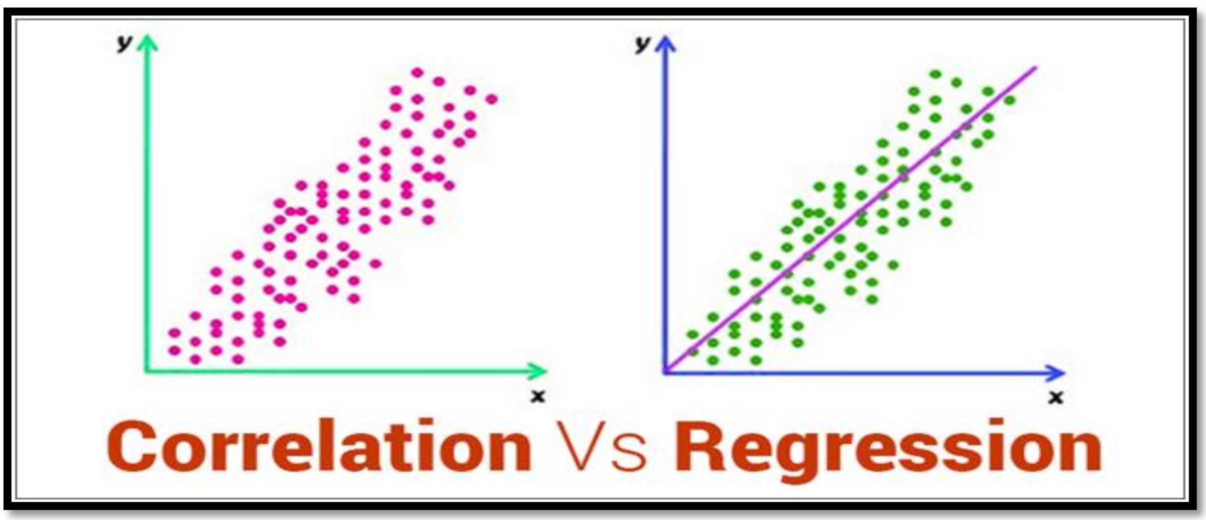
CORRELATION:

Measure the association between two or more continuous variables.

REGRESSION:

The functional relationship between variables

- *SIMPLE LINEAR REGRESSION*
- *MULTIPLE LINEAR REGRESSION*



***Correlation** gives the magnitudinal relationship between the variables whereas **Regression** gives functional relationship between the variables and regression is also used for predicting future values.*

We also learnt about the concepts of correlation, Regression in the direction of applications with R

SIMPLE LINEAR REGRESSION

OBJECTIVE: *The objective is to fit a regression line and find the labor cost for the given labor hours*.

PROCEDURE: *Regression is a functional relationship between two or more variables and is used to predict the future. Here we have only two attributes where "labor hours" is the Independent variable and Labor cost is the dependent variables. Since both the variables are continuous, we use Correlation test. We fit a simple linear Regression line because we have only one Dependent variable and one independent variable.*

PROCESS:

- **Business Objective**

  We are given labor hours and labor cost. We are fitting an equation for labor cost for the given labor hours and predicting the labor cost based on labor hours.

- **Data Understanding**
  1) *Reading the data*
  2) *Checking the dimensions of the data*
  3) *Knowing the structure of the data(data type of the variables)*
  4) *Using summary() we get the descriptive statistics of the data*
  5) *Finding measures of dispersion i.e. the standard deviation*

- **Data Preparation**
  1) *Finding the missing values(if any)*
  2) *Using Boxplot we find the outliers(if any)*
  3) *Removing the outliers by minimizing the values that are more than 95% quantile to 95% and maximizing the values that are less than 5% quantile to 5%*
  4) *Analyzing the data using pairs()*

- **Modeling**
  1) *To find the magnitude relation between the two variables by running correlation test*
  2) *We set a functional relation between the two variables using regression model*

- **Evaluation**
  1) *Evaluating the results*
  2) *Reviewing the process*
  3) *Determining the next steps(if needed)*

- **Deployment**
    1) *Predicting the future*
    2) *Using cbind() we get observed and predicted values in a workbook.*

CODE/OUTPUT:

```
#######################################################

# Setting Working Directory

# Understanding the Business Problem - Predicting labor cost to
the given labor hours.

getwd()

setwd("C:/Users/Sm/Desktop/stanna/Regression/SLR")

getwd()

 [1] "C:/Users/Nav/Desktop/PROJECT/SIMPLE LINEAR REGRESSION"

#######################################################


# Reading data

data=read.csv("vehicle.csv")

data

   lh      lc
1      1.1   66.30
2      2.4  233.03
3      4.2  325.08
4      1.0   66.64
5      4.5  328.66
6      3.1  205.28
7      0.7   49.17
8      2.9  208.80
9      3.4  212.06
10     0.7   44.43
.
.
```

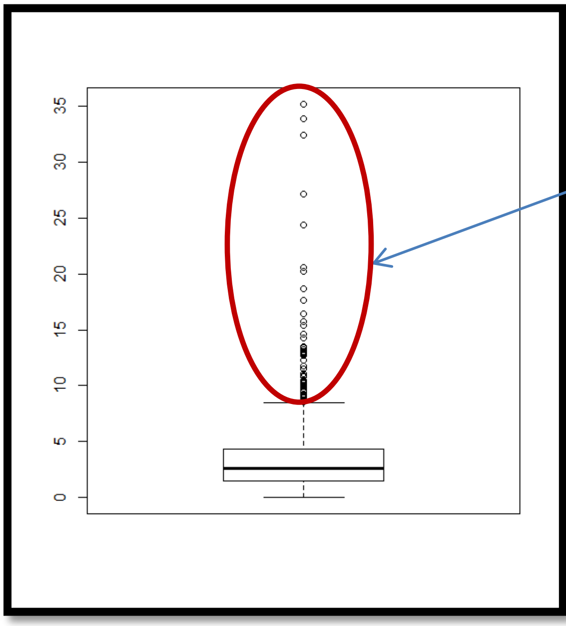
```
.
.
491   5.8  496.38
492   2.7  216.00
493   2.3  142.60
494   1.8  127.25
495   4.8  357.52
496   1.8  139.17
497   1.6  104.16
498   2.1  140.06
499   1.9  140.82
500   5.2  396.39
```

```
########################################################

# Checking the data dimension

#will produce the number of rows and columns of the data.

dim(data)

[1] 1624    2

#Here there are 1624 rows and 2 columns

head(data,10)


#This gives the top 10 values of the data

   lh     lc
1   1.1  66.30
2   2.4 233.03
3   4.2 325.08
4   1.0  66.64
5   4.5 328.66
6   3.1 205.28
7   0.7  49.17
8   2.9 208.80
9   3.4 212.06
10  0.7  44.43
```

```
tail(data,10)

#This gives the bottom 10 values of the data

       lh      lc
1615  3.8  205.14
1616  5.6  356.69
1617  3.9  273.54
1618  0.7   55.78
1619  0.8   78.11
1620 33.9 3234.41
1621  2.1  185.08
1622  4.5  318.10
1623  3.5  306.53
1624  1.8  129.04

#######################################################


# structure of data file-gives the number of varibles in the
data and the scales of the data

str(data)

'data.frame':   1624 obs. of  2 variables:
 $ lh: num  1.1 2.4 4.2 1 4.5 3.1 0.7 2.9 3.4 0.7 ...
 $ lc: num  66.3 233 325.1 66.6 328.7 ...

#######################################################


#Understanding the data with descriptive statistics

summary(data)

lh                lc
 Min.   : 0.000   Min.   :   0.0
 1st Qu.: 1.500   1st Qu.: 106.4
 Median : 2.600   Median : 195.6
 Mean   : 3.308   Mean   : 242.9
 3rd Qu.: 4.300   3rd Qu.: 317.8
 Max.   :35.200   Max.   :3234.4

#######################################################
```

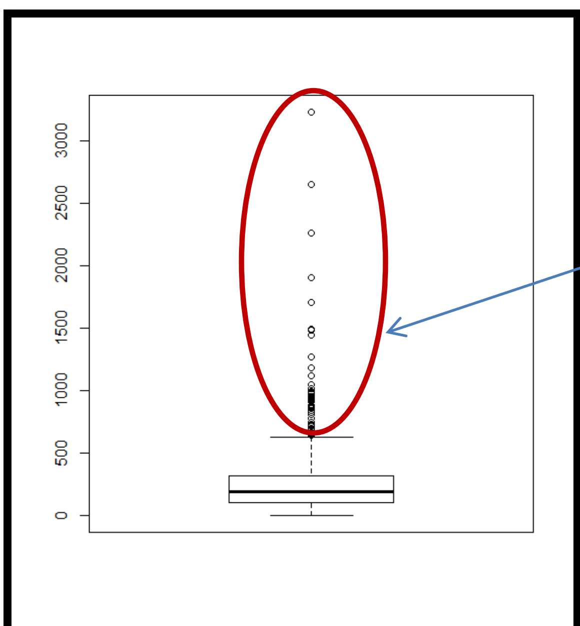
```
# Referring the single variable in the data we use ($)
#finding the standard deviation of the variables lh and lc
s1=sd(data$lh)
s1
[1] 2.912891

s2=sd(data$lc)
s2
[1] 219.5726

######################################################
#To check the Missing values
is.null(data$lh)
[1] FALSE

#FALSE means there are no missing values in the data
is.null(data$lc)
[1] FALSE

#FALSE means there are no missing values in the data
######################################################

# To check the outliers we will use box plot
# Understanding  Descriptive using  Visualisation
```

`Boxplot(data$lh)`



#Here there are outliers

`Boxplot(data$lc)`



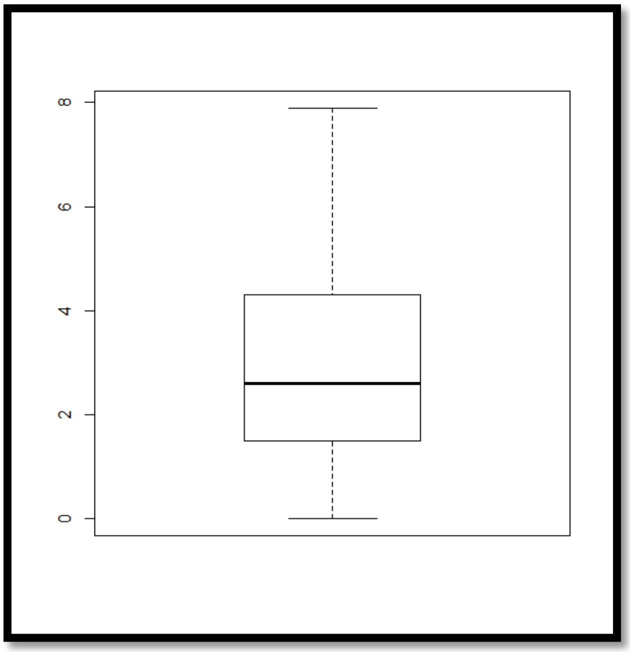
#Here there are outliers

```
xx=data$lh>quantile(data$lh, 0.95)

xx
```

```
  [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
FALSE FALSE FALSE FALSE
  [19] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
  [37] FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE  TRUE FALSE
  [55] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
  [73] FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
  [91] FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
 [109] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
 [127] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
 [145] FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE  TRUE
 [163] FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
 [181] FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
 [199] FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
 [217] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
 [235] FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALS
E FALSE FALSE FALSE FALSE
.
.
.
```

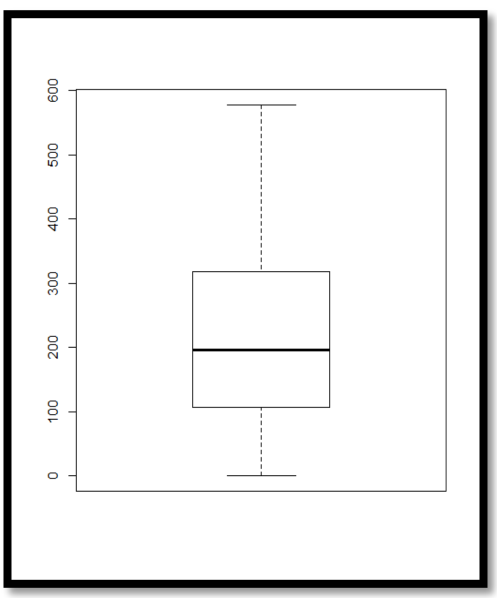
```
#True represents that there is an outlier,false represents there is no outlier

##########################################################


#removing the outliers by reducing the values greater than 95% quantile to 95%

data$lh[data$lh>quantile(data$lh, 0.95)] <- quantile(data$lh, 0.95)
```

```
#checking whether the outliers had been removed or not
boxplot(data$lh)
```



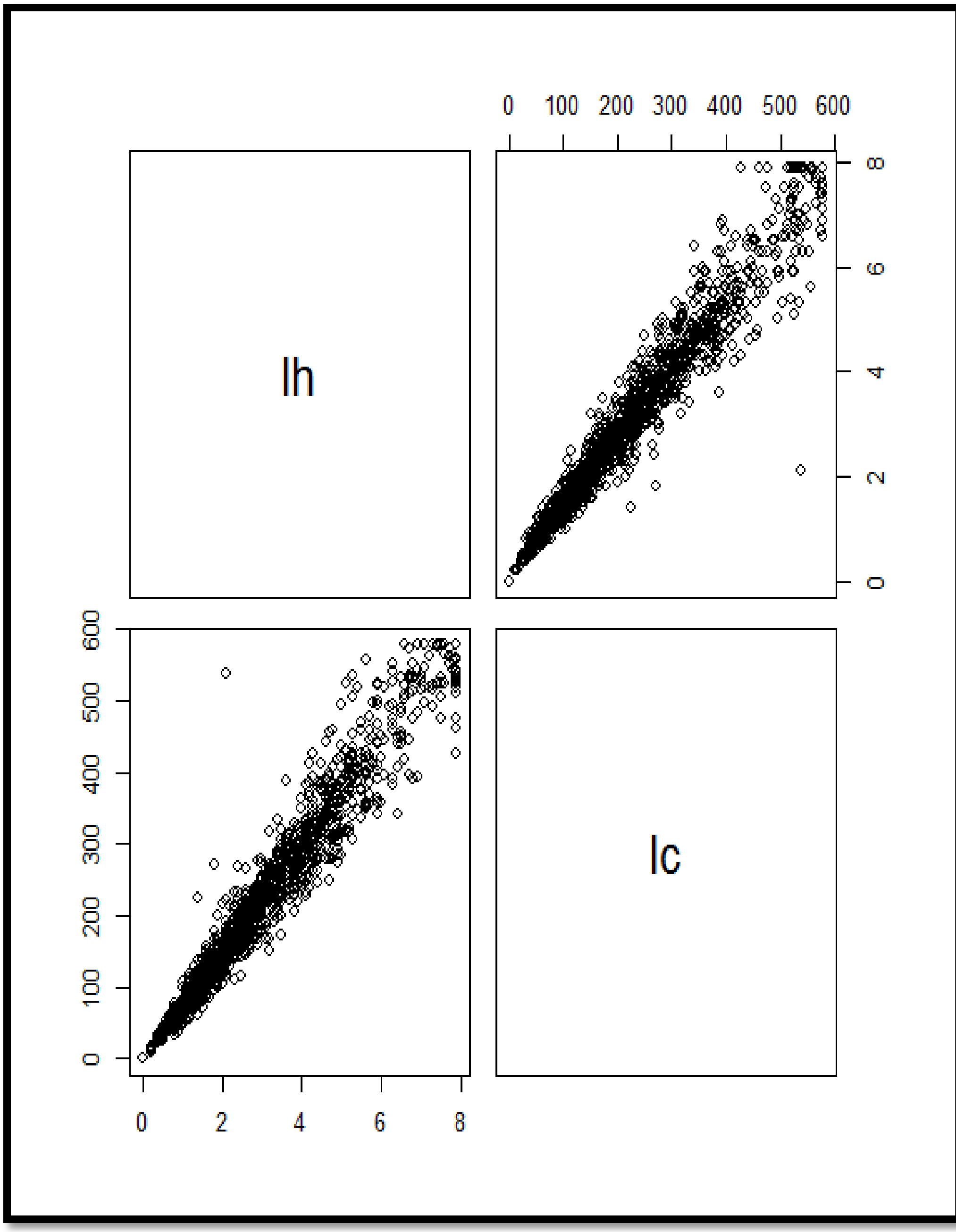
#Here there are no outliers as we removed them

```
data$lc[data$lc>quantile(data$lc, 0.95)] <- quantile(data$lc, 0.95)
boxplot(data$lc)
```

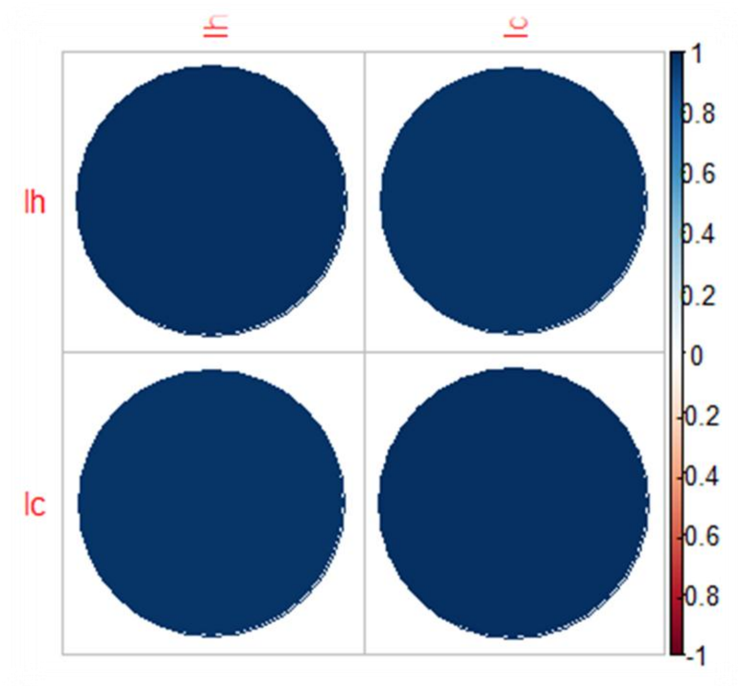


#Here there are no outliers as we removed them

```
###########################################################
#understanding the data using pairs
pairs(data)
```



```
# to  see the correlation
# Checking the correlation between variables
#######################################################
data.cor=cor(data)
library(corrplot)
corrplot(data.cor,method="circle")
```



#There is strong correlation between the variables

```
#######################################################
# Predicting the labor cost based on labor hours
#setting regression model
regmodel <- lm(lc ~ lh ,data=data)
summary(regmodel)
```

```
Call:
lm(formula = lc ~ lh, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-154.21  -13.14   -1.78   11.89  383.34 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.003004   1.392485  -0.002    0.998    
lh          73.427825   0.374482 196.078   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.85 on 1622 degrees of freedom
Multiple R-squared:  0.9595,	Adjusted R-squared:  0.9595 
F-statistic: 3.845e+04 on 1 and 1622 DF,  p-value: < 2.2e-16
#The obtained regression equation is

        Lc=-0.003004+73.427825(lh)

#####################################################

# The equation will be (# lc=-0.0030004+73.427*lh) will be used
for predicting labor cost for the given labor hours

pred=predict(regmodel,data=data)

pred

          1             2             3             4             
5             6             7             8 
 80.767603155 176.223775107 308.393859348  73.424820697 330.4222
06722 227.623252312  51.396473323 212.937687396 
            9            10            11            12            
13            14            15            16 
249.651599685  51.396473323 256.994382143 234.966034770 580.0768
10289 146.852645275 279.022729517 227.623252312 
           17            18            19            20            
21            22            23            24 
 66.082038239 146.852645275 359.793336553 102.795950528 190.9093
40022 168.880992649  14.682561034 301.051076890 
           25            26            27            28            
29            30            31            32 
352.450554095 154.195427733  58.739255781 139.509862817  36.7109
08408 345.107771637 176.223775107 183.566557565 
```

```
           33           34           35           36
37           38           39           40
124.824297902 198.252122480 220.280469854 271.679947059  36.7109
08408  58.739255781 580.076810289  51.396473323
           41           42           43           44
45           46           47           48 ...................................................
```

```
#########################################################

#To write the predicted values into an excel workbook

xx=cbind(data$lc,pred)

write.csv(xx,"predicted.csv")

#########################################################

#predicting the future values and testing the obtained regression equation

newhours = data.frame(lh=c(5.0,6.5))

predict(regmodel, newdata = newhours)
```

```
       1         2
367.1361 477.2779
```

```
#For 5.0 labor hours the labor cost is Rs.367.1361

#For 6.5 labor hours the labor cost is Rs.477.2779
```

MULTIPLE LINEAR REGRESSION

OBJECTIVE: *The objective is to fit a regression line and find the mpg for the given attributes.*

PROCEDURE: *Regression is a functional relationship between two or more variables and is used to predict the future. Here we have many attributes where*

Cyl, Disp, Hp, Drat, Wt, Qsec, Vs, Am, Gear and carb
Are the Independent variables and mpg is the dependent variable. Since the Variables "mpg", "disp", "hp", "drat", "wt","qsec" are continuous, we Use Correlation test and since the variables Am, Vs are categorical we use hypothesis testing by ANOVA. We fit a multiple linear Regression line because we have only one Dependent variable and many independent Variables.

PROCESS:

- **Business Objective**

Predicting Mpg to the given variables (as being multiple independent variable are there hence we are using multiple linear regression)

- **Data Understanding**

1) *Reading the data*
2) *Checking the dimensions of the data*
3) *Knowing the structure of the data(data type of the variables)*
4) *Declaring categorical variables as factors*
5) *Using summary() we get the descriptive statistics of the data*
6) *Finding measures of dispersion i.e. the standard deviation*

- **Data Preparation**

1) *Finding the missing values(if any)*
2) *Using Boxplot we find the outliers(if any)*
3) *Removing the outliers by minimizing the values that are more than 95% quantile to 95% and maximizing the values that are less than 5% quantile to 5%*
4) *Analyzing the data using pairs()*
5) *Subset data to features we wish to keep/use i.e. to find correlation between the continuous variables.*
6) *Applying ANOVA test for the categorical variables*

- **Modeling**

1) *To find the magnitude relation between the two variables by running correlation test*
2) *We set a functional relation between the two variables using regression model*

- **Evaluation**

  1) *Evaluating the results*
  2) *Reviewing the process*
  3) *Determining the next steps(if needed)*
     Using step-wise regression we find the most significant variables to represent the data

- **Deployment**

  1) *Predicting the future*
  2) *Using cbind() we get observed and predicted values in a workbook.*

CODE/OUTPUT:

```
#######################################################

# Understanding the Business Problem - Predicting Mpg to the given variables (as being multiple independent variable are there hence we are using multiple linear regression)

# Setting Working Directory

getwd()

setwd("C:/Users/Sm/Desktop/stanna/Regression/Multiple Regression")

getwd()

 [1] "C:/Users/Nav/Desktop/PROJECT/SIMPLE MULTIPLE REGRESSION"

#######################################################

# Reading data will produce the number of rows and columns of the data.

mtcars=read.csv('mtcars.csv')
```

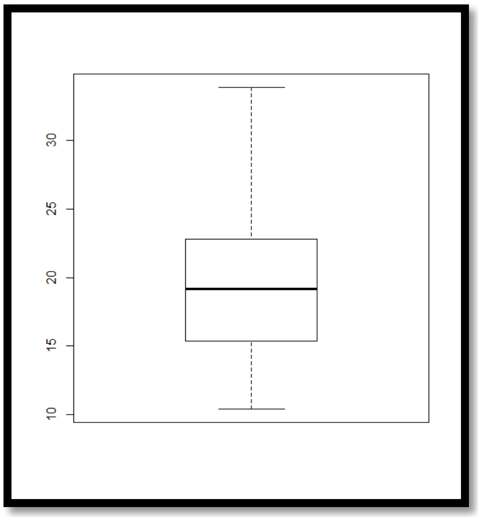
```
dim(mtcars)

 [1] 32 11

#Here there are 32 rows and 11 columns

#####################################################

# structure of data file-gives the number of varibles in the
data and the scales of the data

str(mtcars)

'data.frame':	32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : int  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : int  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : int  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : int  1 1 1 0 0 0 0 0 0 0 ...
 $ gear: int  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: int  4 4 1 1 2 1 4 2 2 4 ...

#####################################################

#declaring the categorical variables as factors and checking the
structure of data to ensure that the categorical variables are
considered as factors or not

mtcars$vs=as.factor(mtcars$vs)

mtcars$am=as.factor(mtcars$am)

str(mtcars)

'data.frame':	32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : int  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : int  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : Factor w/ 2 levels "0","1": 1 1 2 2 1 2 1 2 2 2 ...
 $ am  : Factor w/ 2 levels "0","1": 2 2 2 1 1 1 1 1 1 1 ...
 $ gear: int  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: int  4 4 1 1 2 1 4 2 2 4 ..
```

```
########################################################

#to check whether the data has been read properly or not

head(mtcars)

mpg cyl disp  hp drat    wt  qsec vs am gear carb
1 21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
2 21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
3 22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
4 21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
5 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
6 18.1   6  225 105 2.76 3.460 20.22  1  0    3    1

#Gives the first 6(default) values from the data

########################################################

#Understanding the data with descriptive statistics

summary(mtcars)

mpg             cyl             disp             hp             drat
wt             qsec
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0   Min.   :2
.760   Min.   :1.513   Min.   :14.50
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5   1st Qu.:3
.080   1st Qu.:2.581   1st Qu.:16.89
 Median :19.20   Median :6.000   Median :196.3   Median :123.0   Median :3
.695   Median :3.325   Median :17.71
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :143.8   Mean   :3
.597   Mean   :3.212   Mean   :17.76
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0   3rd Qu.:3
.920   3rd Qu.:3.610   3rd Qu.:18.90
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :253.6   Max.   :4
.930   Max.   :5.293   Max.   :20.10

vs      am          gear            carb
 0:18   0:19   Min.   :3.000   Min.   :1.000
 1:14   1:13   1st Qu.:3.000   1st Qu.:2.000
               Median :4.000   Median :2.000
               Mean   :3.688   Mean   :2.812
               3rd Qu.:4.000   3rd Qu.:4.000
               Max.   :5.000   Max.   :8.000

########################################################
```

```
#Finding the standard deviation of the variables mpg and hp

sd(mtcars$mpg)

[1] 6.026948

sd(mtcars$hp)

[1] 68.56287

#########################################################

# To check the Missing values

is.null(mtcars$mpg)

[1] FALSE

#FALSE means that there are no missing values in the data

#To check the missing values in all the variables of the data

sapply(mtcars, function(df) {

  (sum(is.na(df)==TRUE)/ length(df))*100; })

mpg  cyl disp   hp drat   wt qsec   vs   am gear carb 
   0    0    0    0    0    0    0    0    0    0    0 

#0 means there are no missing values whereas 1  means there are missing values

#########################################################

# To check the outliers we will use box plot

 str(mtcars)

'data.frame':	32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : int  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : int  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
```

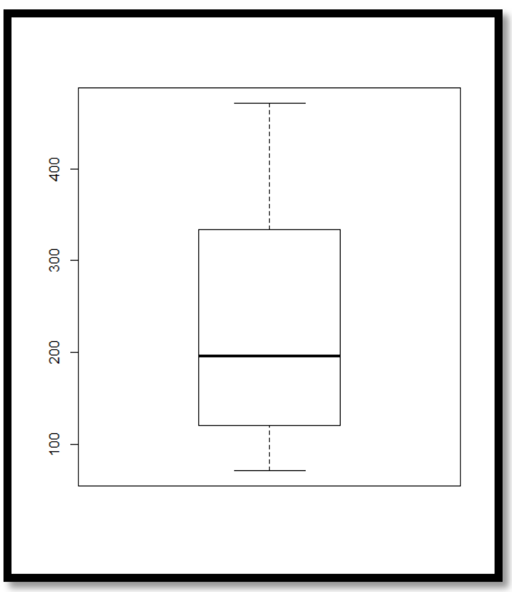
```
 $ vs  : Factor w/ 2 levels "0","1": 1 1 2 2 1 2 1 2 2 2 ...
 $ am  : Factor w/ 2 levels "0","1": 2 2 2 1 1 1 1 1 1 1 ...
 $ gear: int  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: int  4 4 1 1 2 1 4 2 2 4 ...
```

```
# Using boxplots we check the outliers
#Understanding descriptive using Visualization
boxplot(mtcars$mpg)
```



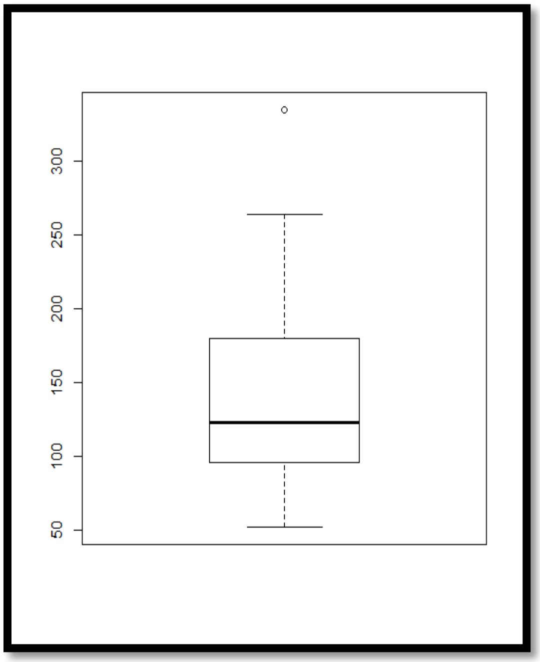
#Here there are no outliers

```
boxplot(mtcars$disp)
```



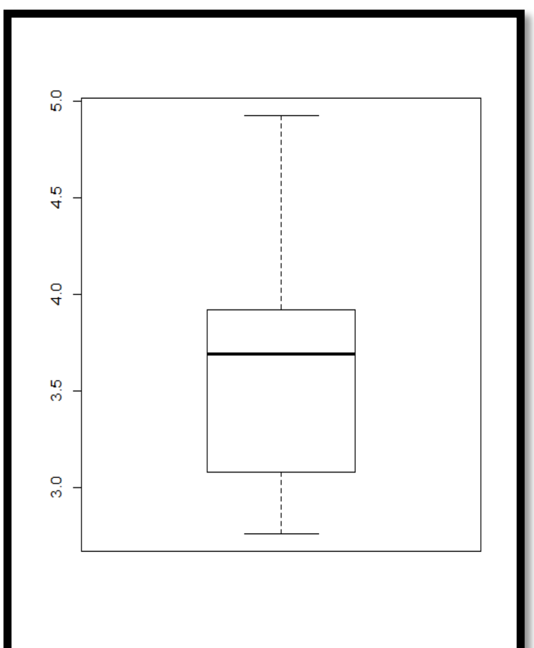
#Here there are no outliers

```
boxplot(mtcars$hp)
```



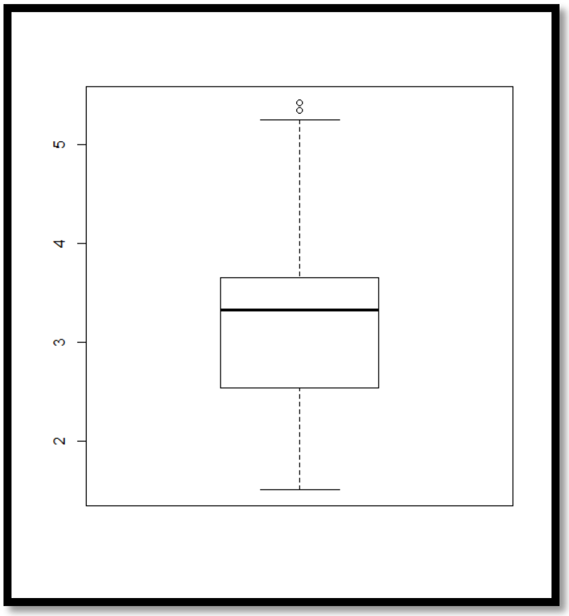
#Here there is an outlier

```
boxplot(mtcars$drat)
```



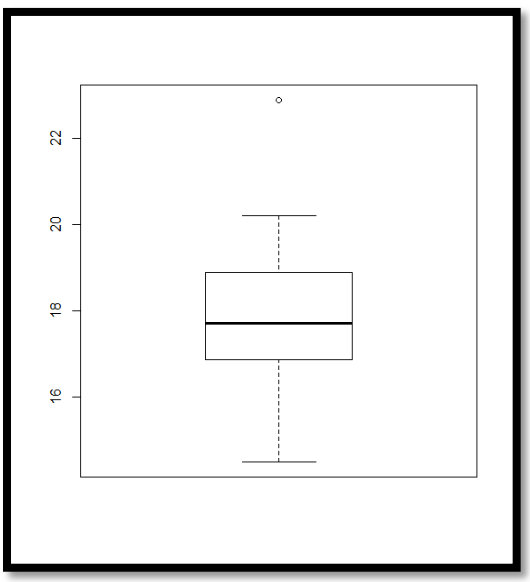
#Here there are no outliers

```
boxplot(mtcars$wt)
```



#Here there are outliers

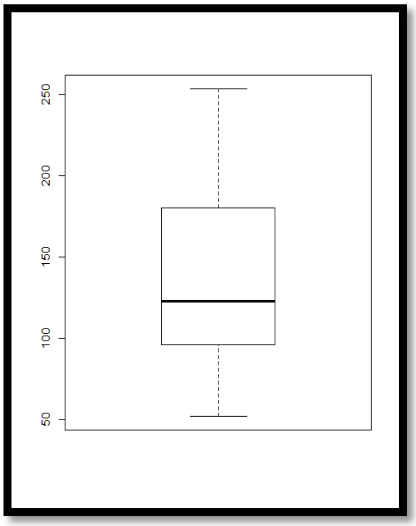
```
boxplot(mtcars$qsec)
```



#Here there is an outliers

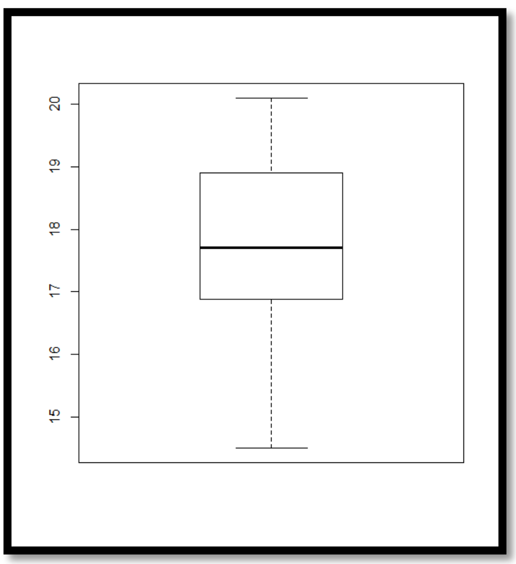
```
#removing the outliers by reducing the values greater than 95% quantile to 95%

mtcars$hp[mtcars$hp>quantile(mtcars$hp, 0.95)] <-
quantile(mtcars$hp, 0.95)
```

```
mtcars$qsec[mtcars$qsec>quantile(mtcars$qsec, 0.95)] <-
quantile(mtcars$qsec, 0.95)

mtcars$wt[mtcars$wt>quantile(mtcars$wt, 0.95)] <-
quantile(mtcars$wt, 0.95)

#checking whether the outliers had been removed or not

boxplot(mtcars$hp)
```



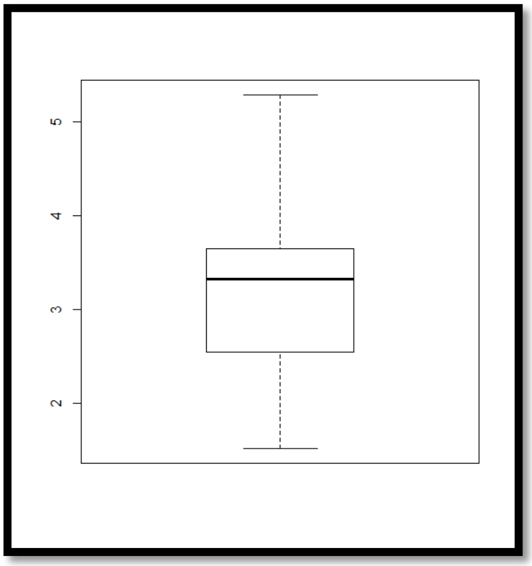
#Here there are no outliers as we removed them

```
boxplot(mtcars$qsec)
```



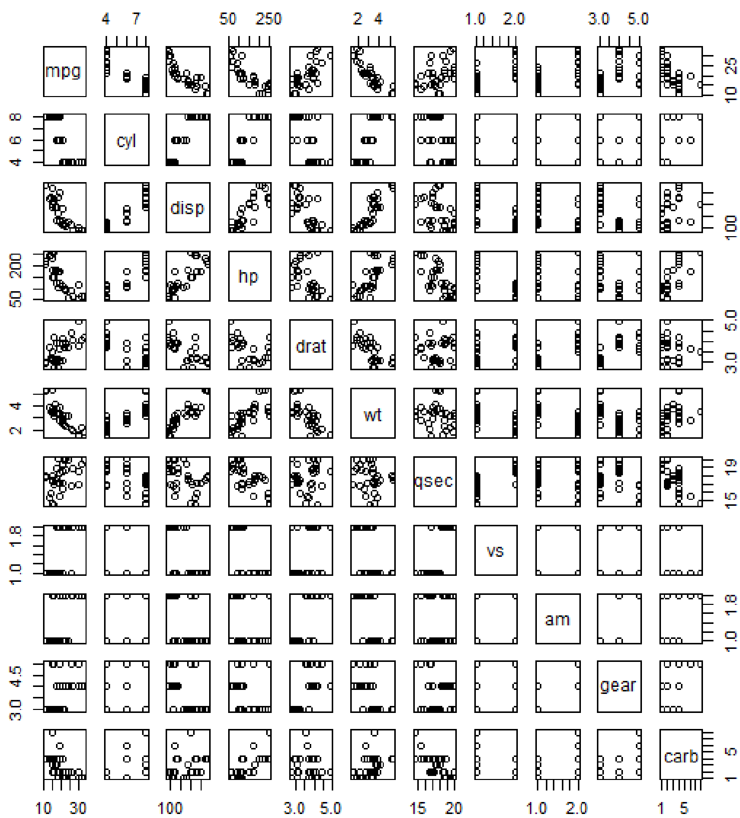
#Here there are no outliers as we removed them

```
boxplot(mtcars$wt)
```



#Here there are no outliers as we removed them

```
##########################################################
#understanding the data using pairs
pairs(mtcars)
```



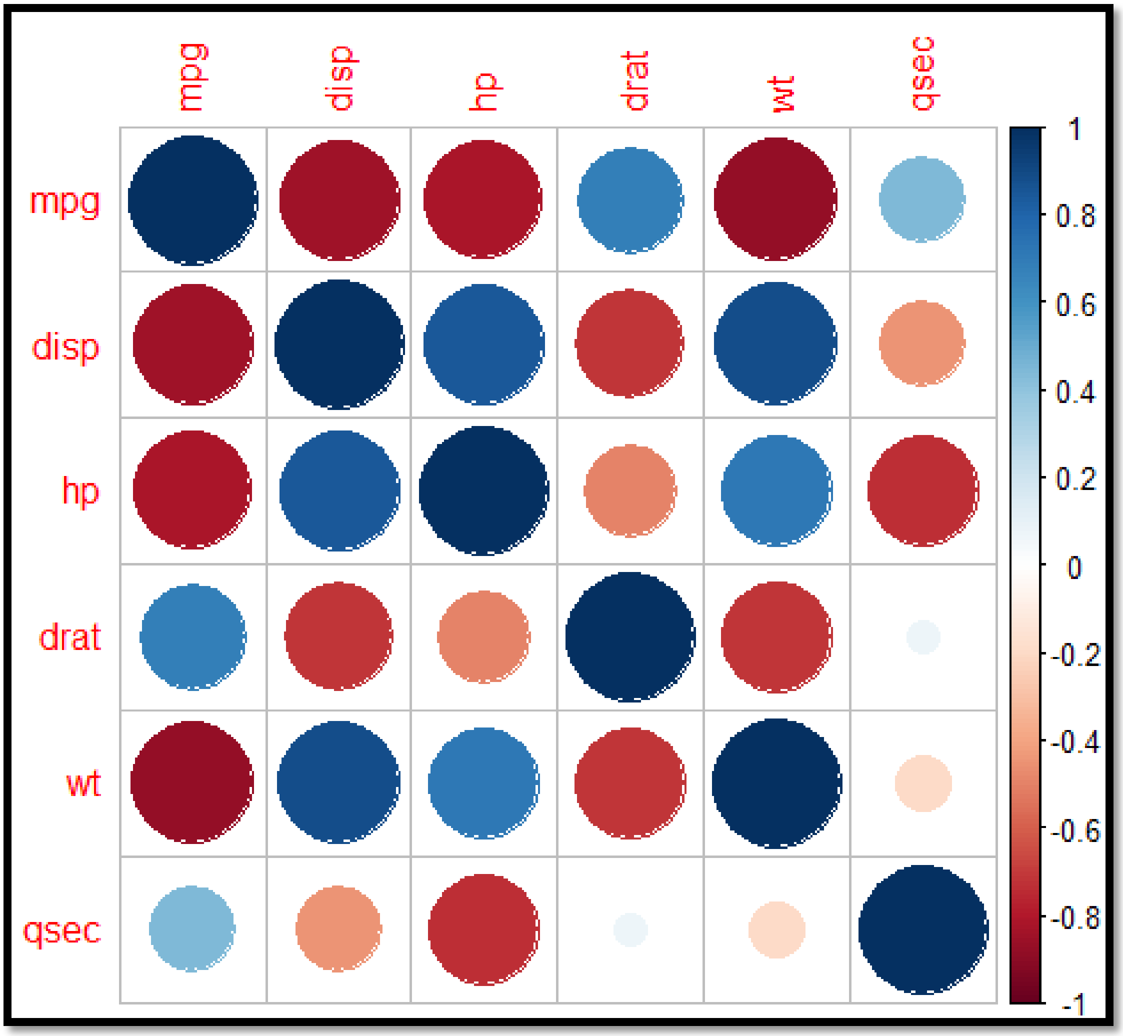
```r
##############################################################

# Subset data to features we wish to keep/use.Finding the
correlation between the continuous variables

features <- c("mpg", "disp", "hp", "drat", "wt","qsec")

data1=mtcars[features]

str(data1)
```

```
'data.frame':	32 obs. of  6 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
```

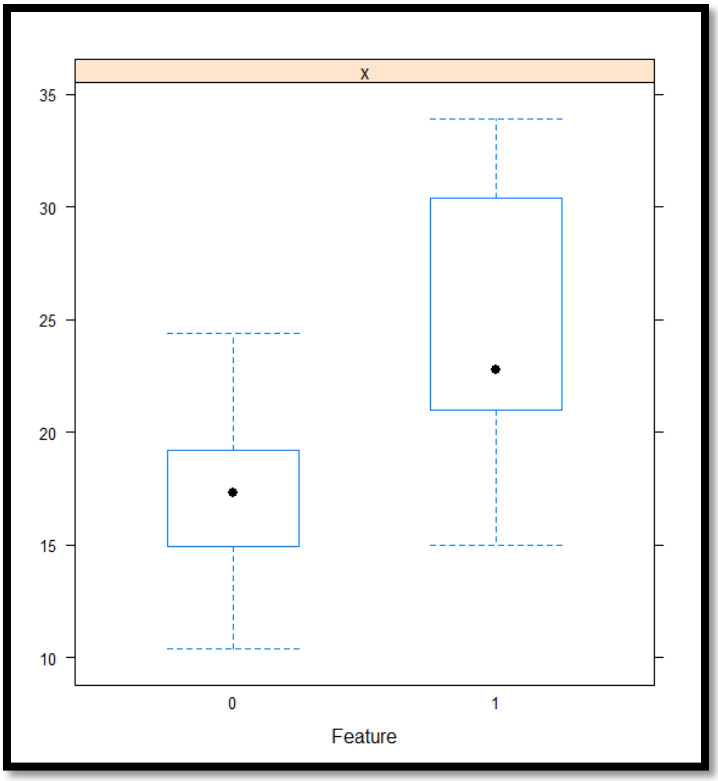
```r
library(corrplot)

#installing corrplot package

pre1.cor = cor(data1)

pre1.cor
```

```
            mpg       disp         hp       drat         wt       qsec
mpg   1.0000000 -0.8475514 -0.8189576  0.6811719 -0.8707197  0.4475638
disp -0.8475514  1.0000000  0.8456704 -0.7102139  0.8888657 -0.4479035
hp   -0.8189576  0.8456704  1.0000000 -0.4982972  0.7158569 -0.7391807
drat  0.6811719 -0.7102139 -0.4982972  1.0000000 -0.7159355  0.0721406
wt   -0.8707197  0.8888657  0.7158569 -0.7159355  1.0000000 -0.1962015
qsec  0.4475638 -0.4479035 -0.7391807  0.0721406 -0.1962015  1.0000000
```

```r
#gives the correlation between the variables

corrplot(pre1.cor,   method="circle")
```

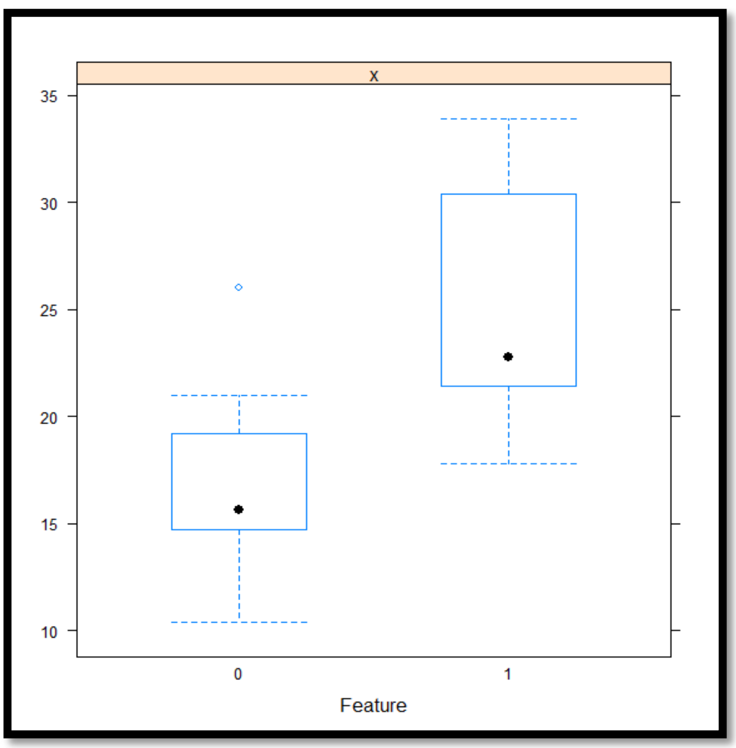



```
#blue circles represent a positive correlation, red circles
represent a negative correlation. The smaller the circles the
lesser the correlation and viceversa


########################################################1

library(caret)

featurePlot(x=mtcars$mpg, y=mtcars$am, plot="box")
```



#Plotting a boxplot to check the relation between mpg and am

```
featurePlot(x=mtcars$mpg, y=mtcars$vs, plot="box")
```



#Plotting a boxplot to check the relation between mpg and vs

```
###########################################################

#Hypothesis testing- applying anova test for one categorical and one continuous variable

str(mtcars)

'data.frame':	32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : int  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : Factor w/ 2 levels "0","1": 1 1 2 2 1 2 1 2 2 2 ...
 $ am  : Factor w/ 2 levels "0","1": 2 2 2 1 1 1 1 1 1 1 ...
 $ gear: int  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: int  4 4 1 1 2 1 4 2 2 4 ...

x1=aov(mtcars$mpg ~ mtcars$vs)

summary(x1)

    Df Sum Sq Mean Sq F value   Pr(>F)
mtcars$vs    1  496.5   496.5   23.66 3.42e-05 ***
Residuals   30  629.5    21.0
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

# p=3.42(e^-0.5) which is less than 0.05and hence we reject the null hypothesis and infer that mpg and vs are related

x2=aov(mtcars$mpg ~ mtcars$am)

summary(x2)

Df Sum Sq Mean Sq F value   Pr(>F)
mtcars$am    1  405.2   405.2   16.86 0.000285 ***
Residuals   30  720.9    24.0
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
# p=0.000285 which is less than 0.05and hence we reject the null hypothesis and infer that mpg and am are related

##########################################################

#setting regression model

mymodel=lm(mpg~.,data=mtcars)

summary(mymodel)

Call:
lm(formula = mpg ~ ., data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.0856 -1.3541 -0.2674  0.9924  4.3785 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept) -9.36344   27.01442  -0.347    0.732  
cyl         -0.05214    0.98019  -0.053    0.958  
disp         0.01697    0.01669   1.017    0.321  
hp          -0.01149    0.02653  -0.433    0.669  
drat         1.21100    1.57366   0.770    0.450  
wt          -4.81095    2.08042  -2.312    0.031 *
qsec         1.90788    1.23562   1.544    0.138  
vs1         -1.58159    2.42612  -0.652    0.522  
am1          1.71827    1.90406   0.902    0.377  
gear         1.37499    1.52771   0.900    0.378  
carb        -0.12307    0.74202  -0.166    0.870  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.543 on 21 degrees of freedom
Multiple R-squared:  0.8794,	Adjusted R-squared:  0.8219 
F-statistic: 15.31 on 10 and 21 DF,  p-value: 1.667e-07

# the regression equation is
Mpg=-9.36344-0.05214(cyl)+ 0.01697(disp)-0.01149(hp)+1.21100(drat)-4.8109   (wt)+1.90788(qsec)-1.58159(vs=1)+1.71827(am=1)+1.37499(gear)-0.12307(carb)

##########################################################
```

```
# Stepwise model to find the best significant variables that
represent the data more precisely

best_model <- step(mymodel, direction = "both")
```

```
Start:  AIC=68.26
mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb

       Df Sum of Sq    RSS    AIC
- cyl   1     0.018 135.85 66.267
- carb  1     0.178 136.01 66.304
- hp    1     1.214 137.05 66.547
- vs    1     2.749 138.58 66.904
- drat  1     3.831 139.66 67.152
- gear  1     5.240 141.07 67.474
- am    1     5.268 141.10 67.480
- disp  1     6.685 142.52 67.800
<none>              135.84 68.262
- qsec  1    15.421 151.26 69.704
- wt    1    34.590 170.43 73.522

Step:  AIC=66.27
mpg ~ disp + hp + drat + wt + qsec + vs + am + gear + carb

       Df Sum of Sq    RSS    AIC
- carb  1     0.232 136.09 64.321
- hp    1     1.255 137.11 64.561
- vs    1     2.773 138.63 64.913
- drat  1     4.451 140.30 65.298
- am    1     5.573 141.43 65.553
- gear  1     6.397 142.25 65.739
- disp  1     7.096 142.95 65.896
<none>              135.85 66.267
- qsec  1    16.260 152.11 67.884
.
.
.
.

Step:  AIC=58.99
mpg ~ wt + qsec + am

       Df Sum of Sq    RSS    AIC
<none>              157.48 58.995
+ disp  1     4.609 152.88 60.045
+ drat  1     2.972 154.51 60.385
```

```
+ carb  1     2.566 154.92 60.469
+ hp    1     1.693 155.79 60.649
+ vs    1     1.453 156.03 60.698
+ gear  1     1.172 156.31 60.756
+ cyl   1     0.474 157.01 60.898
- am    1    25.164 182.65 61.739
- qsec  1   114.742 272.23 74.509
- wt    1   174.875 332.36 80.895
```

#the regression equation after stepwise regression is

Mpg=wt+qsec+am

```
summary(best_model)

Call:
lm(formula = mpg ~ wt + qsec + am, data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.7318 -1.3482 -0.4432  1.4003  4.4543 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   6.2375     7.3807   0.845 0.405209    
wt           -3.9337     0.7055  -5.576 5.76e-06 ***
qsec          1.4253     0.3156   4.517 0.000104 ***
am1           2.8963     1.3693   2.115 0.043444 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.372 on 28 degrees of freedom
Multiple R-squared:  0.8601,	Adjusted R-squared:  0.8452 
F-statistic:  57.4 on 3 and 28 DF,  p-value: 4.426e-12
```

#the obtained regression equation is

Mpg=6.2375-3.9337(wt)+ 1.4253(qsec)+2.8963(am=1)

since the p value is less than 0.05 we reject the null hypothesis and infer that the best fit model represents the data precisely

###

```
#to write the predicted mpg for the given attributes into an excel workbook

pred=predict(best_model,data=mtcars)

xx=cbind(mtcars$mpg,pred)

write.csv(xx,"predicted.csv")

#####################################################

#predicting the future values and testing the obtained regression equation

# Predicting the furture

test=read.csv('test.csv')


test

         X cyl disp  hp drat    wt  qsec vs am gear carb
1          Mazda RX4   6  160 110 3.90 2.620 16.46  0  1    4    4
2      Mazda RX4 Wag   6  160 110 3.90 2.875 17.02  0  1    4    4
3         Datsun 710   4  108  93 3.85 2.320 18.61  1  1    4    1
4     Hornet 4 Drive   6  258 110 3.08 3.215 19.44  1  0    3    1
5  Hornet Sportabout   8  360 175 3.15 3.440 17.02  0  0    3    2


test$am=as.factor(test$am)

str(test)

'data.frame':	5 obs. of  11 variables:
 $ X   : Factor w/ 5 levels "Datsun 710","Hornet 4 Drive",..: 4 5 1 2 3
 $ cyl : int  6 6 4 6 8
 $ disp: int  160 160 108 258 360
 $ hp  : int  110 110 93 110 175
 $ drat: num  3.9 3.9 3.85 3.08 3.15
 $ wt  : num  2.62 2.88 2.32 3.21 3.44
 $ qsec: num  16.5 17 18.6 19.4 17
 $ vs  : int  0 0 1 1 0
 $ am  : Factor w/ 2 levels "0","1": 2 2 2 1 1
 $ gear: int  4 4 4 3 3
 $ carb: int  4 4 1 1 2
```

###

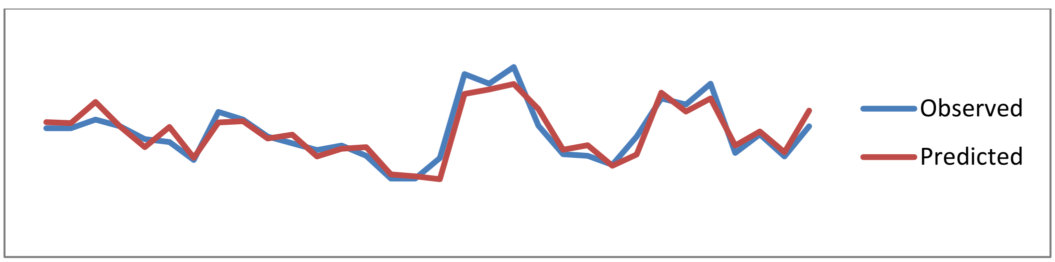
#writing the predicted data which we have tested to an excel workbook

Submitting the results

yy=cbind(test,newpred)

write.csv(yy,"predictedtest.csv")

#Graph for Observed Vs Predicted



###